

Econ 381: Introduction to Econometrics

Handout 5: Simple Linear Regression

Muyang Ren

Department of Economics
University of Tennessee, Knoxville

Roadmap

This handout applies the statistical ideas from Handouts 1–4 to a regression slope.

1. Section 4.1 distinguishes the population line and regression error from the sample line and residual.
2. Section 4.2 derives the ordinary least squares estimator.
3. Section 4.3 introduces R^2 and the standard error of the regression.
4. Section 4.4 explains the least squares assumptions and their role in empirical work.
5. Section 4.5 applies the LLN and CLT to the OLS slope.

The section numbers follow Chapter 4 of Stock and Watson. The running example uses the 420 California school districts in `caschool.dta`.

Section 4.1

The Linear Regression Model

4.1 The empirical question

California policymakers want to know whether smaller classes improve student performance.

How do district test scores vary with the student–teacher ratio?

- ▶ $Y_i = \text{TestScore}_i$: district average test score.
- ▶ $X_i = \text{STR}_i$: enrollment divided by the number of teachers.
- ▶ A straight line gives a simple summary of how the average of Y changes with X .

The sample relationship is an association. A causal interpretation requires assumptions about the other determinants of test scores.

4.1 The population regression line

The target population contains all districts relevant to the superintendent's question, not only the districts that appear in one sample.

$$Y_i = \beta_0 + \beta_1 X_i + u_i.$$

- ▶ $\beta_0 + \beta_1 X$ is the posited **population regression line**.
- ▶ β_1 is the change in average Y associated with a one-unit increase in X , when the model assumptions hold.
- ▶ β_0 is the value of the line at $X = 0$.

What researchers know

The coefficients β_0 and β_1 are fixed population quantities. Researchers and superintendents do not know them because they observe only a sample.

4.1 Interpreting the population coefficients

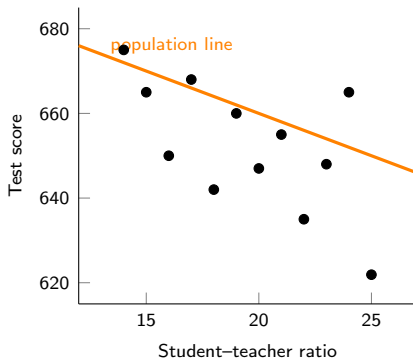
For test scores and the student–teacher ratio,

$$\text{TestScore}_i = \beta_0 + \beta_1 \text{STR}_i + u_i.$$

- ▶ β_1 measures the difference in average test score associated with one more student per teacher.
- ▶ β_0 gives the line's predicted average test score when $\text{STR} = 0$.
- ▶ The intercept has no useful policy interpretation here because no school district has zero students per teacher.

The intercept still matters mathematically because it determines the level of the population line.

4.1 Individual outcomes need not lie on the line



The line describes a proposed population pattern.

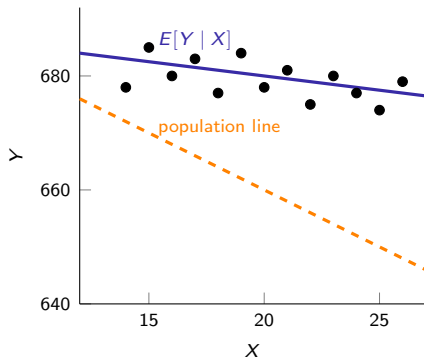
Districts with the same STR can have different test scores because many other factors affect performance.

If observations center around the line at every X ,

$$E[Y | X = x] = \beta_0 + \beta_1 x.$$

This is an assumption about the population. We cannot verify it by estimating the unknown coefficients. Next two cases show how it can fail.

4.1 A linear model with nonzero conditional errors



The equation

$$Y = \beta_0 + \beta_1 X + u$$

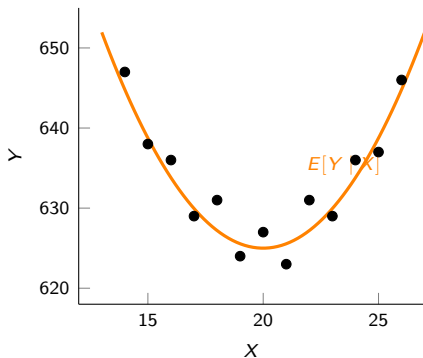
still specifies a linear population model.

Here the observations center around the blue line rather than the posited orange line. Therefore,

$$E[u | X] \neq 0.$$

In this OLS approaches the blue conditional-mean line rather than the population line.

4.1 A nonlinear conditional mean



The observations reveal a curved conditional pattern.

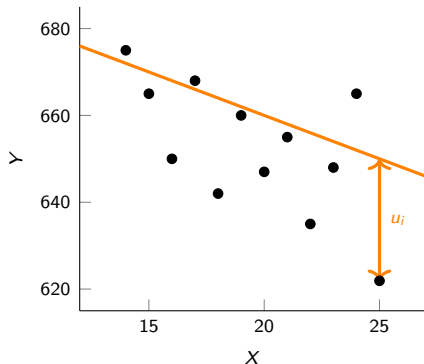
No constants β_0 and β_1 make

$$\beta_0 + \beta_1 X$$

equal to this conditional mean at every X .

A straight line misspecifies $E[Y | X]$. Later chapters introduce nonlinear regression models.

4.1 Regression errors



The **regression error** is

$$u_i = Y_i - (\beta_0 + \beta_1 X_i).$$

It is the vertical distance from unit i to the population line.

It collects other determinants of Y_i and measurement noise in Y_i .

Researchers usually cannot calculate u_i because the population coefficients are unknown.

4.1 The sample regression line and residuals

Researchers use the observed sample to estimate the unknown population coefficients.

Definition

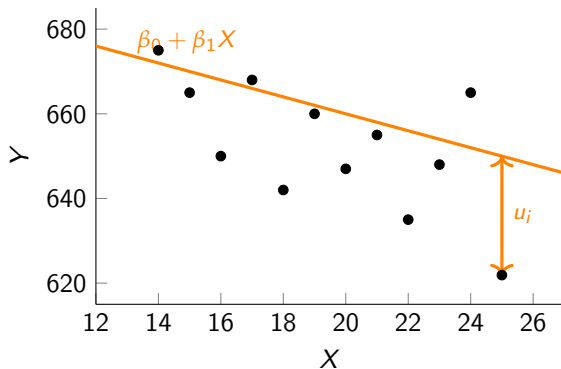
The **sample regression line**, or fitted OLS line, is

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X.$$

At X_i , the point on this line is the **fitted/predicted value** $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$.

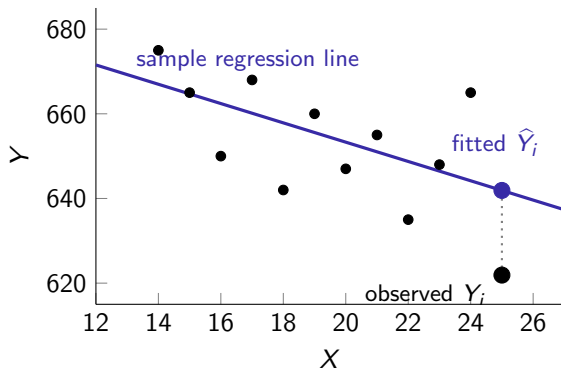
- ▶ Before sampling, $\hat{\beta}_0$ and $\hat{\beta}_1$ are random because the sample is unknown.
- ▶ After sampling, researchers calculate both estimates from the observed data.
- ▶ The vertical gap between Y_i and $\hat{Y}_i$ is the **residual**: $\hat{u}_i = Y_i - \hat{Y}_i$.

4.1 Population line and regression error



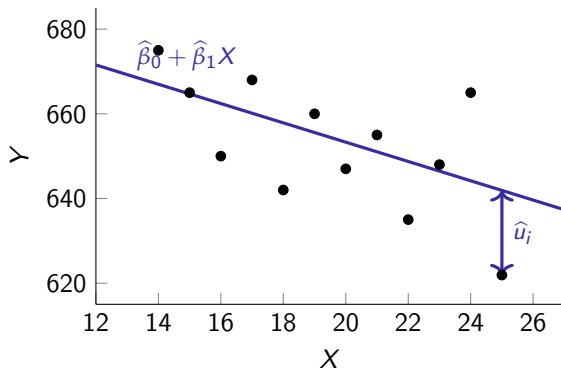
The population line and u_i describe the underlying model. Researchers do not observe them directly.

4.1 Observed and fitted values



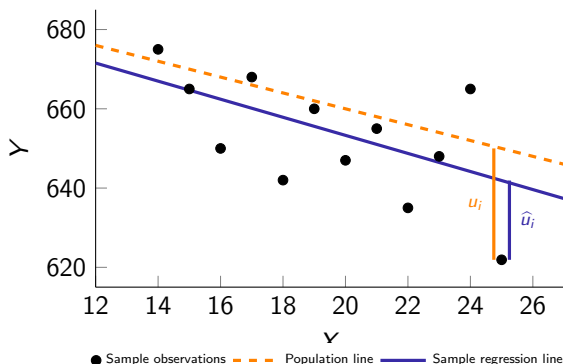
At $X_i = 25$, the black point is the observed outcome and the purple point is the predicted value. Their vertical difference is the residual.

4.1 Sample line and residual



Researchers compute the sample line and its residuals to approximate the unknown population relationship.

4.1 Regression error and residual for the same district



For $STR_i = 25$ and $TestScore_i = 621.9$, the residual is computable:
 $\hat{u}_i = 621.9 - (698.9 - 2.28 \times 25) = -20.0$. The regression error remains unknown.

4.1 Model fit and estimation error

The residual contains the population error and the error from estimating the coefficients:

$$\hat{u}_i = u_i - (\hat{\beta}_0 - \beta_0) - (\hat{\beta}_1 - \beta_1)X_i.$$

Model fit

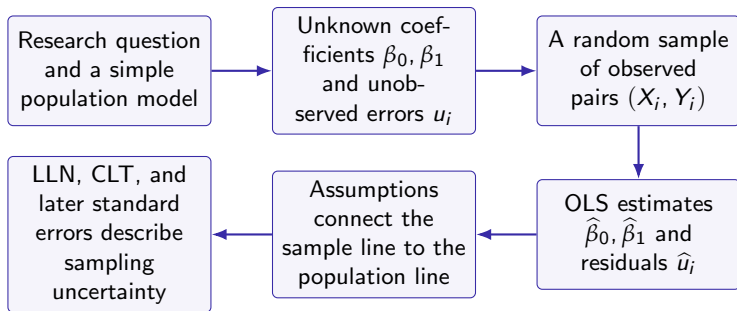
The regression errors u_i describe how individual outcomes vary around the population line. Even a linear conditional-mean model need not fit every observation.

Estimation accuracy

Sampling makes the estimated coefficients differ from their population values. Under suitable assumptions, this difference shrinks as the sample grows.

Researchers observe residuals $\hat{u}_i$, not regression errors u_i . Small residuals alone cannot prove that $\hat{\beta}_0$ and $\hat{\beta}_1$ are close to their population values.

4.1 The empirical workflow



The additional assumptions

Compared to Handout 4, regression requires additional assumptions about the error and model specification to connect the sample line to the intended population line.

Section 4.2

Estimating the Coefficients of the Linear Regression Model

4.2 The motivation for OLS

The population line is unknown, so researchers need a rule for constructing a sample line from the data.

- ▶ A fitted line produces one residual for every observation.
- ▶ Residuals can be positive or negative. Researchers want to avoid large misses in either direction.
- ▶ Simply adding residuals would make positive and negative values to cancel.

Least squares idea

Square every residual, add the squares, and choose the line that makes this sum as small as possible.

This rule produces the **ordinary least squares**, or OLS, sample regression line.

4.2 The OLS minimization problem

For any candidate intercept b_0 and slope b_1 , define

$$Q(b_0, b_1) = \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2.$$

OLS chooses

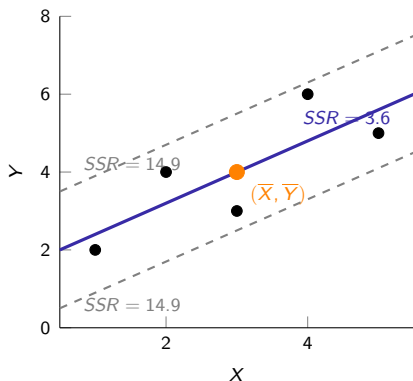
$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{b_0, b_1} Q(b_0, b_1).$$

Multivariable calculus gives two first-order conditions:

$$\begin{aligned}\frac{\partial Q}{\partial b_0} &= -2 \sum_{i=1}^n (Y_i - b_0 - b_1 X_i) = 0, \\ \frac{\partial Q}{\partial b_1} &= -2 \sum_{i=1}^n X_i (Y_i - b_0 - b_1 X_i) = 0.\end{aligned}$$

We will solve the same problem with a one-variable approach.

4.2 Choosing the intercept for a fixed slope



All three candidate lines have slope $b_1 = 0.8$. Only the intercept changes.

The upper line produces mostly negative residuals. The lower line produces mostly positive residuals.

The middle line balances the residuals at zero and has the smallest sum of squared residuals.

For a fixed slope

$$\text{OLS} \implies \sum_{i=1}^n \hat{u}_i = 0$$

4.2 The OLS line passes through the sample means

For a candidate line, write its residual as

$$\hat{u}_i = Y_i - b_0 - b_1 X_i.$$

$$\bar{\hat{u}} = \frac{1}{n} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)$$

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For a candidate line, write its residual as

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$$\begin{aligned}\bar{\hat{u}} &= \frac{1}{n} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i) \\ &= \bar{Y} - b_0 - b_1 \bar{X}.\end{aligned}$$

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At the best vertical position, positive and negative residuals balance:

$$\bar{\hat{u}} = \frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0.$$

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At the best vertical position, positive and negative residuals balance:

$$\bar{\hat{u}} = \frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0.$$

Set $\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X}$ equal to zero and solve for the intercept:

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.}$$

Therefore, $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$, so the OLS line passes through $(\bar{X}, \bar{Y})$.

4.2 De-meaning creates a one-variable problem

Define deviations from the sample means:

$$\tilde{X}_i = X_i - \bar{X}, \quad \tilde{Y}_i = Y_i - \bar{Y}.$$

Subtract $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$ from $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i$:

$$\tilde{Y}_i = \hat{\beta}_1 \tilde{X}_i + \hat{u}_i.$$

The remaining OLS problem

Centering removes the intercept. The OLS slope $\hat{\beta}_1$ minimizes the sum of squared residuals from regressing $\tilde{Y}_i$ on $\tilde{X}_i$:

$$\hat{\beta}_1 = \arg \min_{b_1} \sum_{i=1}^n (\tilde{Y}_i - b_1 \tilde{X}_i)^2.$$

The slope compares whether above-average X values occur with above- or below-average Y values.

4.2 The OLS slope from one derivative

Differentiate the centered objective with respect to b_1 :

$$Q'(b_1) = -2 \sum_{i=1}^n \tilde{X}_i (\tilde{Y}_i - b_1 \tilde{X}_i).$$

At the minimum, $Q'(\hat{\beta}_1) = 0$, so

$$\sum_{i=1}^n \tilde{X}_i \tilde{Y}_i = \hat{\beta}_1 \sum_{i=1}^n \tilde{X}_i^2.$$

Hence,

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}} = \frac{s_{XY}}{s_X^2}.$$

4.2 The centered objective is a convex quadratic

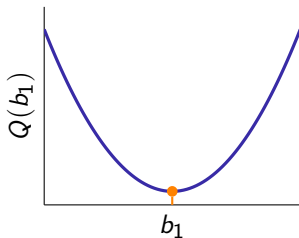
Alternatively, expand the objective:

$$Q(b_1) = b_1^2 \sum \tilde{X}_i^2 - 2b_1 \sum \tilde{X}_i \tilde{Y}_i + \sum \tilde{Y}_i^2.$$

Because $\sum \tilde{X}_i^2 > 0$, this quadratic opens upward.

$$\hat{\beta}_1 = \frac{\sum \tilde{X}_i \tilde{Y}_i}{\sum \tilde{X}_i^2}$$

is the unique point at the bottom of the curve.



4.2 Recovering the intercept

The previous step gives $\hat{\beta}_1$. The sample line must also pass through $(\bar{X}, \bar{Y})$:

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}.$$

Solve this equation for the intercept:

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.}$$

1. De-mean X and Y .
2. Estimate the slope from the centered variables.
3. Use the sample means to recover the intercept.

4.2 A three-observation OLS example

i	X_i	Y_i	$\tilde{X}_i$	$\tilde{Y}_i$	$\tilde{X}_i \tilde{Y}_i$	$\hat{Y}_i$	$\hat{u}_i$
1	1	2					
2	2	3					
3	3	7					

Step 1: observed data

The sample contains $(X_i, Y_i) = (1, 2), (2, 3), (3, 7)$.

4.2 A three-observation OLS example

i	X_i	Y_i	$\tilde{X}_i$	$\tilde{Y}_i$	$\tilde{X}_i \tilde{Y}_i$	$\hat{Y}_i$	$\hat{u}_i$
1	1	2	-1	-2			
2	2	3	0	-1			
3	3	7	1	3			

Step 2: sample means and centered values

$\bar{X} = 2$ and $\bar{Y} = 4$. Subtract the corresponding mean from every observation.

4.2 A three-observation OLS example

i	X_i	Y_i	$\tilde{X}_i$	$\tilde{Y}_i$	$\tilde{X}_i \tilde{Y}_i$	$\hat{Y}_i$	$\hat{u}_i$
1	1	2	-1	-2	2		
2	2	3	0	-1	0		
3	3	7	1	3	3		

Step 3: slope and intercept

$$\hat{\beta}_1 = \frac{2 + 0 + 3}{(-1)^2 + 0^2 + 1^2} = 2.5, \quad \hat{\beta}_0 = 4 - 2.5(2) = -1.$$

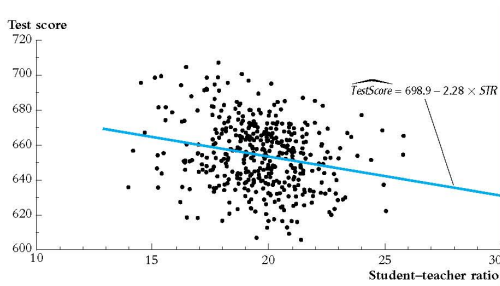
4.2 A three-observation OLS example

i	X_i	Y_i	$\tilde{X}_i$	$\tilde{Y}_i$	$\tilde{X}_i \tilde{Y}_i$	$\hat{Y}_i$	$\hat{u}_i$
1	1	2	-1	-2	2	1.5	0.5
2	2	3	0	-1	0	4.0	-1.0
3	3	7	1	3	3	6.5	0.5

Step 4: fitted values and residuals

$\hat{Y}_i = -1 + 2.5X_i$. The residuals are 0.5, -1, 0.5, and $\sum_i \hat{u}_i = 0$.

4.2 OLS for California schools



$$\widehat{TestScore} = 698.9 - 2.28 STR$$

- ▶ $\hat{\beta}_1 = -2.28$
test-score points per student per teacher.
- ▶ $\hat{\beta}_0 = 698.9$
extrapolates to $STR = 0$ and has little substantive meaning.
- ▶ The line summarizes the sample association.

4.2 Interpretation of the estimated slope

$$\widehat{TestScore} = 698.9 - 2.28 STR.$$

Interpretation: Districts with one more student per teacher have test scores that are 2.28 points lower on average in this sample.

- ▶ The sign gives the direction of the association.
- ▶ The magnitude uses test-score points per unit of STR.
- ▶ A causal interpretation requires the other determinants of test scores to be unrelated to STR, or held fixed by the research design.

Why is the slope estimate not yet causal?

District income, English learner share, and school resources may affect both STR and test scores.

4.2 Stata commands

- ▶ Estimate OLS with an intercept:

```
regress testscr str
```

- ▶ Plot observations and the fitted line:

```
twoway (scatter testscr str) (lfit testscr str)
```

- ▶ Save fitted values and residuals:

```
predict testscr_hat    predict u_hat, residuals
```

The data remain `caschool.dta`.

Section 4.3

Measures of Fit and Prediction Accuracy

4.3 Why the slope does not describe fit

The slope gives the direction and magnitude of the fitted relationship. It does not tell us how well the fitted values approximate the observed data.

Relative fit

What fraction of the variance in Y does the fitted line capture?

Use R^2 .

Residual size

How far are observations typically from their fitted values, in the units of Y ?

Use the SER.

R^2 is unit-free and lies between zero and one. The SER has the same units as Y .

4.3 Two prediction rules

Suppose we want to predict each observed Y_i .

Mean-only prediction

Predict the same value $\bar{Y}$ for every observation. The total squared prediction error is

$$TSS = \sum_i (Y_i - \bar{Y})^2.$$

Regression prediction

Use the fitted value $\hat{Y}_i$ for each observation. The remaining squared prediction error is

$$RSS = \sum_i (Y_i - \hat{Y}_i)^2.$$

Because the horizontal mean line is one candidate OLS could choose, the fitted regression cannot have a larger squared error: $RSS \leq TSS$. The reduction

$$ESS = TSS - RSS$$

is the variation captured by the sample regression line.

4.3 TSS, ESS, and RSS

Quantity	Formula	Meaning
TSS	$\sum_i (Y_i - \bar{Y})^2$	Squared misses from predicting every observation by $\bar{Y}$
ESS	$\sum_i (\hat{Y}_i - \bar{Y})^2$	Reduction in squared misses achieved by the sample line
RSS	$\sum_i (Y_i - \hat{Y}_i)^2 = \sum_i \hat{u}_i^2$	Squared misses remaining after fitting the line

For OLS with an intercept,

$$TSS = ESS + RSS.$$

Total sample variation in Y equals the part captured by the line plus the part left in the residuals.

4.3 The definition and endpoints of R^2

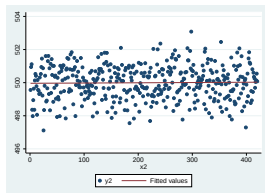
$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}, \quad 0 \leq R^2 \leq 1.$$

Assume that Y varies in the sample, so $TSS > 0$.

- ▶ If $R^2 = 0$, then $ESS = 0$. Every fitted value equals $\bar{Y}$, and the sample line is horizontal with slope zero.
- ▶ If $0 < R^2 < 1$, the fitted line captures part of the variation in Y .
- ▶ If $R^2 = 1$, then $RSS = 0$. Every fitted value equals the observed value, so the sample line passes through every data point.

In a regression with one regressor and an intercept, R^2 also equals the squared sample correlation between X and Y .

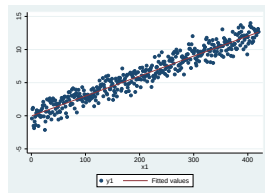
4.3 Different degrees of sample fit



$$R^2 = 0$$



$$R^2 = 0.05$$



$$R^2 = 0.9$$

A value near one means that observations lie close to the fitted line. Exactly $R^2 = 1$ requires every observation to lie on that line.

4.3 The standard error of the regression

Residuals capture the imperfect fit from regressions. Their variability therefore gives another measure of fit.

$$SER = \sqrt{\frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2}.$$

- ▶ The SER resembles the sample standard deviation of the residuals.
- ▶ It reports the typical residual size in the units of Y .
- ▶ Technical note: The divisor $n - 2$ accounts for estimating the intercept and slope.

Stata terminology

With one regressor and an intercept, Stata's Root MSE equals the SER.

4.3 Fit measures in Stata

```
regress testscr str, robust
```

Regression with robust standard errors

Number of obs = 420
F(1, 418) = 19.26
Prob > F = 0.0000
R-squared = 0.0512
Root MSE = 18.581

testscr		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]

str		-2.279808	.5194892	-4.39	0.000	-3.300945 -1.258671
_cons		698.933	10.36436	67.44	0.000	678.5602 719.3057

$$\widehat{TestScore} = 698.9 - 2.28 \cdot STR$$

Stata reports $R^2 = 0.0512$ and Root MSE= 18.581. The coefficient table reports a different standard error for each estimated coefficient.

4.3 Questions about the SER and R^2

Assume $TSS > 0$.

1. If $SER = 0$, what is R^2 ?
2. Does a small SER imply that R^2 is close to one?
3. Does a large SER imply that R^2 is close to zero?

4.3 Questions about the SER and R^2

Assume $TSS > 0$.

1. If $SER = 0$, what is R^2 ? **Answer:** $SER = 0$ implies $RSS = 0$, so $R^2 = 1$.
2. Does a small SER imply that R^2 is close to one? **Answer:** No. R^2 compares RSS with TSS , while the numerical size of the SER depends on the units of Y .
3. Does a large SER imply that R^2 is close to zero? **Answer:** No, for the same reason. Measuring dollars instead of thousands of dollars changes the SER but leaves R^2 unchanged.

4.3 Fit of the California regression

$$\widehat{TestScore} = 698.9 - 2.28 STR, \quad R^2 = 0.051, \quad SER = 18.6.$$

- ▶ STR accounts for about 5.1% of the sample variation in test scores.
- ▶ District scores typically lie about 18.6 points from the fitted line.
- ▶ A low R^2 does not show that class size lacks policy importance.

Fit and causality

R^2 and SER describe how the sample line fits observed outcomes. Neither measure determines whether the slope has a causal interpretation.

4.3 Two different standard errors

	Standard error of regression	Standard error of $\hat{\beta}_1$
Notation	SER or Stata Root MSE	$SE(\hat{\beta}_1)$
What varies?	Outcomes around the fitted line	Estimated slopes across repeated samples
Units	Units of Y	Units of Y per unit of X
Main use	Describe sample prediction error	Construct confidence intervals and tests for β_1

Section 4.5 introduces the sampling distribution behind $SE(\hat{\beta}_1)$. The next handout explains how to estimate this standard error and use it for inference.

Section 4.4

The Least Squares Assumptions

4.4 Why assumptions matter

OLS produces a fitted line whenever the sample contains variation in X .

Assumptions determine what researchers may learn from that line.

- ▶ Is X systematically related to omitted determinants of Y ?
- ▶ Does the sample contain independent and representative observations from the target population?
- ▶ Can a few extreme observations dominate the result?

The practical role of assumptions

The assumptions justify using the OLS sample slope to learn about a population slope. They also support a causal interpretation and valid measures of sampling uncertainty.

Regression output cannot verify these assumptions. Researchers justify them using the data collection process, institutional knowledge, plots, and robustness checks.

4.4 The three least squares assumptions

Assumption	Plain-language meaning	Why it matters
A1. $E[u \mid X = x] = 0$	Omitted determinants of Y average to zero at every value of X	Makes the population line equal the conditional average and rules out systematic omitted factors related to X
A2. (X_i, Y_i) are i.i.d.	Observations come independently from the same population distribution	Supports the LLN and CLT for the estimator
A3. Large outliers are unlikely	X and Y do not have “unusually large” observations	Limits the influence of extreme observations and supports large-sample approximations

4.4 A1 is the key condition for the slope target

The **most important** condition for estimating the population slope is

$$E[u \mid X = x] = 0.$$

At every value of X , positive and negative regression errors must average to zero. Then

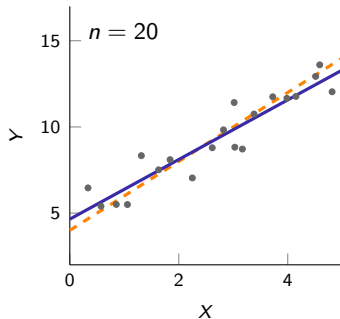
$$E[Y \mid X = x] = \beta_0 + \beta_1 x.$$

- ▶ Observations may remain widely scattered around the line.
- ▶ The assumption concerns their average vertical position at each X .
- ▶ If the average error changes with X , OLS converges to a different slope even in a very large sample.

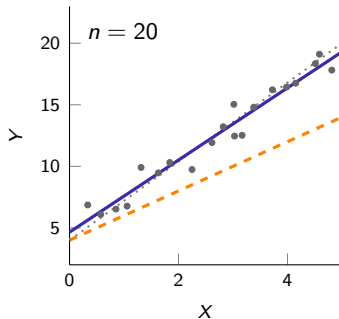
4.4 A1 and growing samples

The two simulations use the same X values and random noise. The right side adds the systematic component $1.2X$ to the regression error.

A1 holds: $E[u | X] = 0$



A1 fails: $E[u | X] \neq 0$



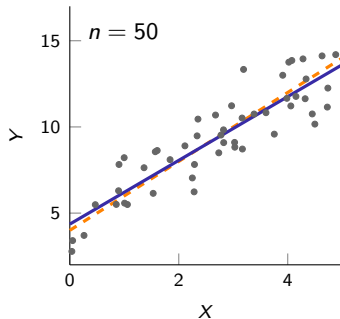
Dashed orange: population line. Solid blue: sample line. Dotted gray: conditional average when A1 fails.

With $n = 20$, both sample lines contain substantial sampling noise.

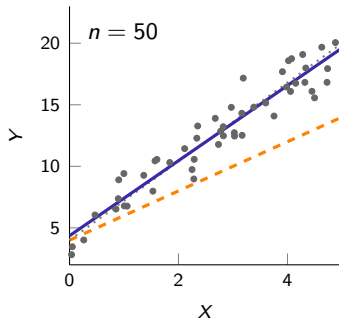
4.4 A1 and growing samples

The two simulations use the same X values and random noise. The right side adds the systematic component $1.2X$ to the regression error.

A1 holds: $E[u | X] = 0$



A1 fails: $E[u | X] \neq 0$



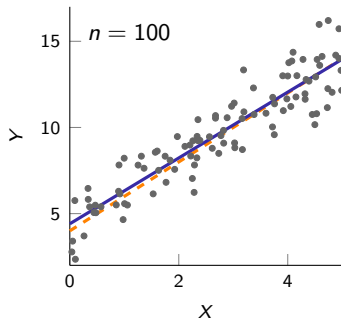
Dashed orange: population line. Solid blue: sample line. Dotted gray: conditional average when A1 fails.

With $n = 50$, the sample lines begin to stabilize.

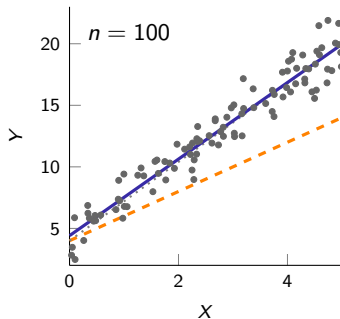
4.4 A1 and growing samples

The two simulations use the same X values and random noise. The right side adds the systematic component $1.2X$ to the regression error.

A1 holds: $E[u | X] = 0$



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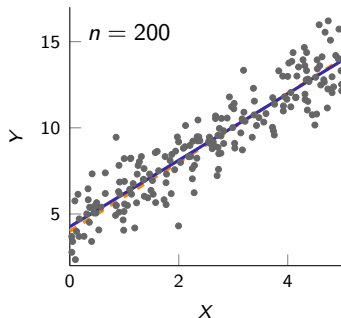
Dashed orange: population line. Solid blue: sample line. Dotted gray: conditional average when A1 fails.

With $n = 100$, the distinction becomes clearer.

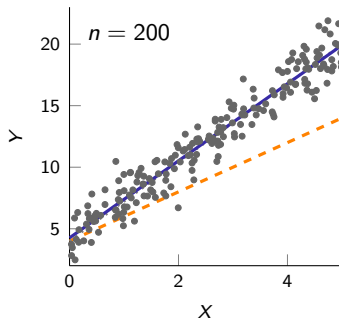
4.4 A1 and growing samples

The two simulations use the same X values and random noise. The right side adds the systematic component $1.2X$ to the regression error.

A1 holds: $E[u | X] = 0$



A1 fails: $E[u | X] \neq 0$



Dashed orange: population line. Solid blue: sample line. Dotted gray: conditional average when A1 fails.

When A1 holds, the sample line approaches the population line. When A1 fails, it approaches the conditional average instead.

4.4 A1 and causal interpretation

Researchers often want β_1 to represent the causal effect of X on Y . In the school model, u_i contains other determinants of test scores.

- ▶ A1 requires the average of these other factors to remain constant across values of STR.
- ▶ Researchers do not observe u_i , β_0 , or β_1 . They therefore cannot test A1 directly from the regression residuals.
- ▶ Researchers must ask whether any omitted determinant of test scores is also related to STR.

Plausible omitted factors

District income and the percentage of English learners may affect test scores and may also be related to class size.

Later classes introduce control variables and instrumental variables that address these concerns.

4.4 Random assignment as a benchmark for A1

Imagine that a computer randomly assigns students to different class sizes.

- ▶ STR varies for reasons unrelated to family background, prior achievement, or teacher quality.
- ▶ Economists call this **exogenous variation** because it does not depend on other determinants of the outcome.
- ▶ Random assignment therefore makes $E[u \mid STR] = 0$ credible by design.

Observational school data

Fully randomized experiments are rare. Districts choose staffing and families choose locations, so researchers must ask whether observed STR is related to omitted determinants of test scores. Later methods add controls or use external sources of plausibly exogenous variation.

4.4 What independent and identically distributed mean

A2 says the sampled pairs (X_i, Y_i) are independent and identically distributed.

Independent observations

The district selected in one draw does not affect the district selected in another draw. Knowing one district's STR and test score does not help us predict another district's STR or test score in the future draws.

Identically distributed observations

Every observation is drawn from the same population using the same sampling method. Districts may have different STRs and test scores, but each district in the population has the same chance of being selected.

Simple random sampling

Drawing district names independently from the same population gives i.i.d. observations.

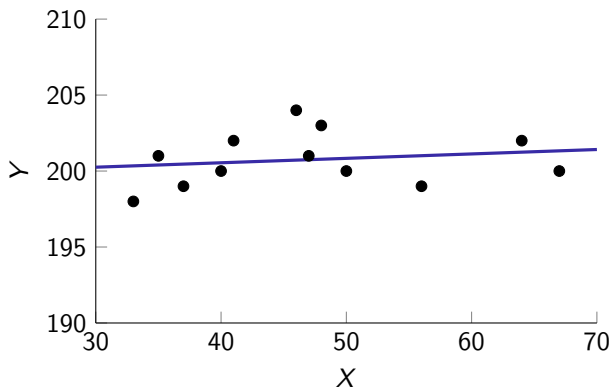
4.4 When the i.i.d. assumption fails

- ▶ **Clustered data:** Districts in the same county may share policies or local economic shocks. After observing one district, we learn something about nearby districts, so the observations are not independent.
- ▶ **Time data:** STR and test scores for one county across years may share a trend. Current enrollment plan may depend on previous years, so observations across time are not independent.

Consequence

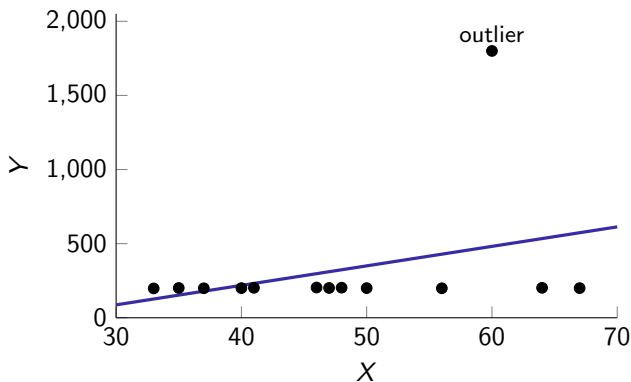
The usual i.i.d. LLN and CLT no longer apply directly. Later methods adjust standard errors or model this dependence.

4.4 A3 and the influence of an outlier



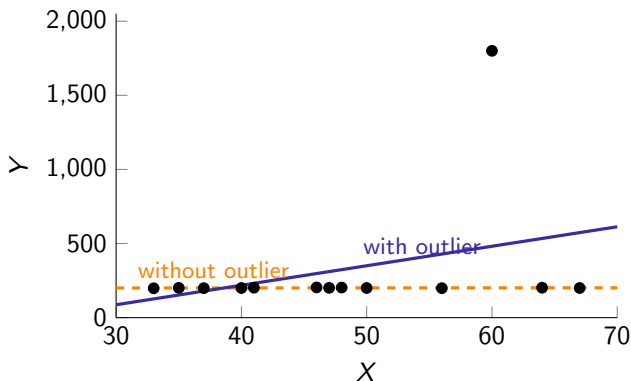
Without the outlier, the sample slope is approximately 0.03.

4.4 A3 and the influence of an outlier



Adding one extreme observation changes the sample slope to approximately 13.15.

4.4 A3 and the influence of an outlier



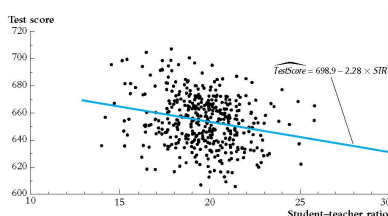
An extreme X value has high leverage and can rotate the fitted line.

Implication: Researchers should check units, coding, and whether the observation belongs to the target population.

Section 4.5

The Sampling Distribution of the OLS Estimators

4.5 A fixed population target



Consider an experiment, treat the 420 districts in `caschool.dta` as a fixed population. Let $X = STR$ and $Y = TestScore$. Its full-data fitted slope is

$$\beta_1^{pop} = -2.28.$$

A1 in this experiment

Under this population β_1^{pop} , test scores are centered around this line at each STR. Positive and negative regression errors therefore average to zero, so A1 holds.

The scatterplot alone cannot verify A1. We assume $\beta_1^{pop} = -2.28$ so that A1 holds, so we can focus on sampling uncertainty in the OLS estimates.

4.5 The repeated-sampling rule

The population data and target slope remain fixed. What changes from sample to sample is the set of districts observed.

1. Each student draws $n = 50$ district names with replacement.
2. Each student estimates one OLS slope.
3. Different samples produce different estimates.

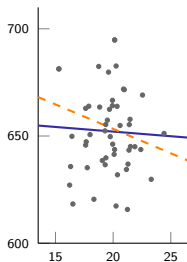
Why the estimated slope is random

Each student's sample is random. Because $\hat{\beta}_1$ is calculated from the 50 districts in that student's sample, the value of $\hat{\beta}_1$ changes when the sampled districts change.

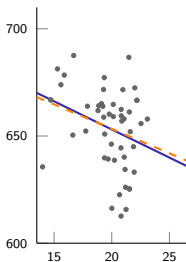
This mirrors Handout 4: the population mean was fixed, while the sample mean changed across samples.

4.5 Three samples, three estimated lines

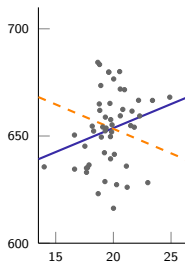
Sample A: $\hat{\beta}_1 = -0.43$



Sample B: $\hat{\beta}_1 = -2.63$



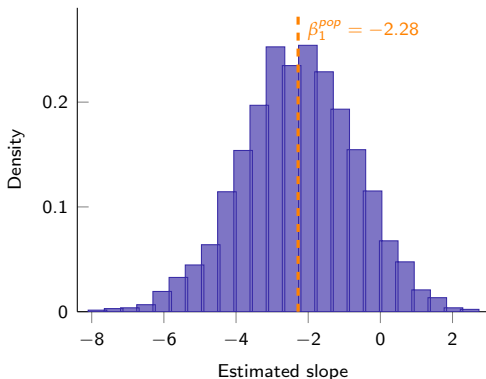
Sample C: $\hat{\beta}_1 = 2.24$



Solid blue: sample line. Dashed orange: fixed population target.

Every student uses the same estimator and sample size. Their estimates differ because their samples contain different districts.

4.5 The sampling distribution of $\hat{\beta}_1$



Three thousand students each draw 50 districts with replacement and report one slope.

The resulting sampling distribution has

$$\begin{aligned}\text{mean} &\approx -2.28, \\ \text{SD} &\approx 1.57.\end{aligned}$$

The center lies near the fixed population target, while the spread measures sampling uncertainty.

4.5 The slope estimator parallels the sample mean

	Mean problem	Regression problem
Fixed target	$\mu_Y = E[Y]$	β_1
Random estimator	$\bar{Y}$	$\hat{\beta}_1$
LLN result	$\bar{Y} \xrightarrow{p} \mu_Y$	$\hat{\beta}_1 \xrightarrow{p} \beta_1$
CLT result	Standardized $\bar{Y}$ is approximately normal	Standardized $\hat{\beta}_1$ is approximately normal

The next handout develops $SE(\hat{\beta}_1)$ and uses it to construct confidence intervals and hypothesis tests. We first need to see how the LLN and CLT enter the regression problem.

4.5 The exact slope estimation error

Substitute $Y_i = \beta_0 + \beta_1 X_i + u_i$ into the OLS slope formula:

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (X_i - \bar{X}) u_i}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Multiply the numerator and denominator by $1/n$:

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}) u_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}.$$

By the LLN, the denominator becomes close to

$$\text{Var}(X)$$

in a large sample.

4.5 The first-order expansion

Rewrite the numerator around the population mean $E[X]$:

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}) u_i = \frac{1}{n} \sum_{i=1}^n (X_i - E[X]) u_i - (\bar{X} - E[X]) \bar{u}.$$

Both $\bar{X} - E[X]$ and $\bar{u}$ become very small as the sample grows, so their product is negligible relative to the sample-average term.

Thus the first-order expansion is

$$\hat{\beta}_1 - \beta_1 = \frac{1}{\text{Var}(X)} \underbrace{\left[\frac{1}{n} \sum_{i=1}^n (X_i - E[X]) u_i \right]}_{\text{a sample average}} + \text{negligible errors}.$$

The complicated estimator now looks like a scaled sample average plus a negligible remainder.

4.5 The sample averages to zero

Define

$$v_i = (X_i - E[X])u_i.$$

Why does A1 imply that v_i averages to zero?

- ▶ Among units with the same $X = x$, the multiplier $x - E[X]$ is the same fixed number.
- ▶ A1 says the errors u_i average to zero within that group.
- ▶ Multiplying a zero group average by $x - E[X]$ still gives zero.
- ▶ Combining all values of X therefore gives

$$E[v_i] = E[(X_i - E[X])u_i] = 0.$$

4.5 The LLN gives consistency

A2 and A3 allow the LLN to turn the sample average into its population mean:

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i \xrightarrow{p} 0.$$

This implies

$$\hat{\beta}_1 - \beta_1 \approx \frac{1}{\text{Var}(X)} \cdot \bar{v} \xrightarrow{p} 0.$$

Hence,

$$\boxed{\hat{\beta}_1 \xrightarrow{p} \beta_1.}$$

Implication: With a larger sample, our estimator $\hat{\beta}_1$ will get closer to the population value β_1 .

4.5 The CLT gives approximate normality

For $v_i = (X_i - E[X])u_i$, A1 gives $E[v_i] = 0$. A2 and A3 also support the CLT:

$$\frac{\bar{v}}{\sqrt{\text{Var}(v_i)/n}} \xrightarrow{d} N(0, 1).$$

Since $\hat{\beta}_1 - \beta_1 \approx \bar{v} / \text{Var}(X)$,

$$\boxed{\frac{\text{Var}(X)(\hat{\beta}_1 - \beta_1)}{\sqrt{\text{Var}(v_i)/n}} \xrightarrow{d} N(0, 1).}$$

Equivalently, in a large sample,

$$\hat{\beta}_1 \overset{\text{approx.}}{\sim} N\left(\beta_1, \frac{\text{Var}(v_i)}{n[\text{Var}(X)]^2}\right).$$

4.5 What determines slope precision?

The large-sample variance has the form

$$\text{Var}(\hat{\beta}_1) \approx \frac{1}{n} \frac{\text{Var}[(X_i - E[X])u_i]}{[\text{Var}(X)]^2}.$$

- ▶ A larger sample generally lowers the standard deviation of the estimator at the $1/\sqrt{n}$ rate.
- ▶ More variation in X supplies more information about the slope.
- ▶ Larger and more variable regression errors make the slope less precise.

4.5 What determines slope precision?

The large-sample variance has the form

$$\text{Var}(\hat{\beta}_1) \approx \frac{1}{n} \frac{\text{Var}[(X_i - E[X])u_i]}{[\text{Var}(X)]^2}.$$

Change	Effect on $\text{Var}(\hat{\beta}_1)$	Intuition
Larger n	Decreases	More observations provide more information about the slope.
Larger $\text{Var}(X)$	Decreases	More variation in X makes the slope easier to identify.
Larger $\text{Var}(u)$	Increases	Outcomes are more dispersed around the population line.

Next step

These population quantities are unknown. The next handout explains how to estimate the sampling uncertainty using $SE(\hat{\beta}_1)$.

New notions and theory in Handout 5

New notions

- ▶ Population line: $\beta_0 + \beta_1 X$
- ▶ Regression error:
 $u_i = Y_i - \beta_0 - \beta_1 X_i$
- ▶ Sample line: $\hat{\beta}_0 + \hat{\beta}_1 X$
- ▶ Residual: $\hat{u}_i = Y_i - \hat{Y}_i$
- ▶ Fit: R^2 and SER
- ▶ Sampling distribution of $\hat{\beta}_1$

New theory

- ▶ De-mean first:

$$\hat{\beta}_1 = \frac{\sum \tilde{X}_i \tilde{Y}_i}{\sum \tilde{X}_i^2},$$
$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

- ▶ A1–A3 support consistency and large-sample normality.
- ▶ We turn $\hat{\beta}_1 - \beta_1$ into a scaled sample average.
- ▶ The LLN gives consistency. The CLT gives approximate normality.

The population line is fixed. The estimated line and residuals depend on the realized sample. The next handout develops standard errors, confidence intervals, and tests.